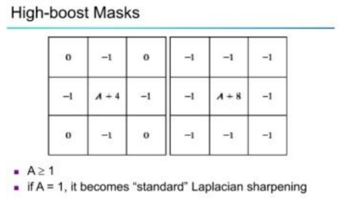
24. Perform Sharpening of Image using High-Boost Masks.



AIM:

Sharpening of Image using High-Boost Masks.

PROGRAM:

```
import cv2 import
numpy as np

def high_boost_filter(image, boost_factor):
    kernel_size = 3

    # Create averaging kernel     kernel = np.ones((kernel_size, kernel_size),
dtype=np.float32) / (kernel_size ** 2)

    # Blur the image     blur_image =
cv2.filter2D(image, -1, kernel)

    # Apply High-Boost Filtering     mask = image +
(image - blur_image) * boost_factor

    # Convert to 8-bit image     mask =
np.clip(mask, 0, 255).astype(np.uint8)

    return mask

image_path = r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png"

image = cv2.imread(image_path)

if image is None:
    print("Error: Image not found!") else:
    # Apply High-Boost Filter     sharpened_image =
high_boost_filter(image, 1.5)

    # Save the output image     cv2.imwrite(
r"C:\Users\Susmitha\Downloads\ITA0510 Lab\sharpened_image.jpg",
sharpened_image
```

```
    )

    # Display the images     cv2.imshow("Original Image", image)
cv2.imshow("High-Boost Sharpened Image", sharpened_image)

    cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT:

OUTPUT:

RESULT:
Thus the program is executed successfully.